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Developing a method to estimate building height from Sentinel-1 data

Xuecao Li[a], Yuyu Zhou[a,*], Peng Gong[b], Karen C. Seto[c], Nicholas Clinton[d]

[a] Department of Geological and Atmospheric Sciences, Iowa State University, Ames, IA 50011, USA
[b] Department of Earth System Science, Tsinghua University, Beijing 100084, China
[c] School of Forestry and Environmental Studies, Yale University, New Haven, CT 06511, United States
[d] Google, Inc., 1600 Amphitheatre Pkwy, Mountain View, CA 94043, USA

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ABSTRACT

Building height is one of the key variables in studying anthropogenic activities and built environments in the urban system. However, the information of urban building height over large areas is still limited due to the lack of available data. In this study, we developed an approach to estimate building height using the Sentinel-1 Ground Range Detected data. First, we proposed an indicator of VVH that integrates the dual-polarization information (i.e., VV and VH) from the Sentinel-1 synthetic aperture radar data. Second, we developed a building height model using the indicator of VVH and the reference building height from airborne light detection and ranging (LiDAR) images in seven cities in the United States (US). Third, we estimated building height of major cities with area larger than 500 km^2 in the US using the developed model. The root mean square error between estimated and reference height is 1.5 m in seven study cities in the US. The comparison with the Laser observations from the Ice, Cloud, and land Elevation Satellite data in available footprints also indicates that the estimated building height from Sentinel-1 data is reliable. The estimated building height from Sentinel-1 data shows a better agreement with the reference data compared to the Advanced Land Observing Satellite data in seven US cities. Our results of building height in major US cities indicate that there are more cities with tall buildings in the Northeastern US compared to the Western and Southern US, with more aggregated tall buildings in central business districts of cities. The information of building height from this study is of great value in urban studies, such as estimating energy consumption and carbon emissions in urban areas.

1. Introduction

Information on building height is important for understanding the impacts of the built environment on the environment and human well-being. Urban structure and form, including the three-dimensional (3-D) layout of buildings, have been correlated with energy consumption, greenhouse gas emissions, and the urban heat island (UHI) effect. According to the latest Intergovernmental Panel on Climate Change (IPCC) report, urban areas emit more than 70% global greenhouse gas emissions from final energy use, with urban form and structure being a leading factor (Seto et al., 2014). As a complement to urban two-dimensional (2-D) information (Li et al., 2015), the third dimension such as building height is helpful to more accurately estimate the distribution of population (Gong et al., 2011; Li and Zhou, 2017), building energy use (Güneralp et al., 2017; Li et al., 2017a), and emissions in urban areas (Gurney et al., 2009). Also, the vertical structure of buildings provides additional information about embodied carbon in commodities such as steel, cement, and wood that are embedded in buildings (Hutyra et al., 2014; Xi et al., 2016), and this information is crucial for global carbon budget studies. Additionally, buildings with different heights and distributions alter the pathway of wind, which further affects the transmission of pollutions and heat fluxes caused by the UHI (Cárdenas Rodríguez et al., 2016; Clinton and Gong, 2013; Sobstyl et al., 2018; Wang et al., 2018).

Despite its importance for scientific studies, there is limited information on building height at regional and global scales. There have been a number of efforts in estimating the 3-D building height, but most of these studies have been limited to local scales, with no data at the global scale. Conventional optical remote sensing with medium or moderate resolution is inadequate to extract the 3-D building height information. Instead, high resolution satellite image, interferometric synthetic aperture radar (InSAR), and airborne Light Detection and Ranging (LiDAR), are natural sources for extracting height information over urban areas (Frolking et al., 2013; Gong et al., 2011; Gong et al., 2002; Takaku et al., 2016; Wang et al., 2014). However, these data are difficult to collect at the global scale because of the cost of data

* Corresponding author.
E-mail address: yuyuzhou@iastate.edu (Y. Zhou).

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acquisition and subsequent processing, particularly for less developed countries (Gong et al., 2011; Li and Gong, 2016).

Existing building height studies using satellite data are limited regarding the spatial resolution and coverage. Frolking et al. (2013) and Mahendra and Seto (2019) measured the built-up infrastructure at the global scale using backscatters from NASA's SeaWinds microwave scatterometer. However, the resolution of SeaWinds is 0.05°, which is too coarse for many urban studies. Gong et al. (2011) estimated building height in Beijing using the Geoscience Laser Altimeter System (GLAS) onboard NASA Ice, Cloud, and land Elevation Satellite (ICESat). However, the ICESat data only partially cover the land surface with limited observation duration. In addition to these globally distributed satellite data, most building height studies were carried out at the city scale, using airborne LiDAR data, high resolution stereo images, or InSAR (Bonczak and Kontokosta, 2019; Hao et al., 2016; Sun et al., 2017; Thiele et al., 2007; Xu et al., 2015; Yu et al., 2010). Nevertheless, the acquisition and processing of such data are difficult over large areas.

To fill this knowledge gap on urban height estimation over large areas, we developed a building height model using the Sentinel-1 data and generated a height dataset of major cities in the US. The remainder of this paper describes the datasets used in this study (Section 2), the building height model (Section 3), the results with discussion (Section 4), and concluding remarks (Section 5).

2. Datasets

The data used in this study include the Sentinel-1 Ground Range Detected (GRD), the reference building height, the ICESat, the Shuttle Radar Topography Mission (SRTM), and the Advanced Land Observing Satellite (ALOS). The Sentinel-1 GRD is the major data source in estimating building height. The GRD data were derived from a dual-polarization C-band synthetic aperture radar (SAR) instrument on the Sentinel-1 satellite (Torres et al., 2012). The signal recorded in the GRD data is the backscatter coefficient that measures the incident microwave radiation scattered by the radiated terrain. The scattering behavior depends on the geometry of terrain elements and their electromagnetic characteristics (e.g., materials) (Koppel et al., 2017). Given that there are different combinations of instrument mode and polarization in the Sentinel-1 data, we created a homogeneous GRD subset by selecting GRD scenes with a dual polarization (i.e., VV and VH) from the instrument mode of interferometric wide swath. To mitigate the impact of trees in calculating the building height, we masked non-urban areas using the urban extent map from the global human settlement layer (Pesaresi et al., 2016).

The reference building height data in seven US cities were collected and used as references in developing the building height model (Fig. 1). The reference data with a finer spatial resolution than 10 m were mainly derived from airborne LiDAR data through publicly accessible sources (Table S1). The reference LiDAR-derived height was aggregated to moderate resolutions (e.g., 500 m) when developing the building height model. Trees were not included in the reference building height data. In general, there is a notable decline in building height from the urban center to surrounding suburban and rural areas, and such a pattern varies across different cities with a wide range of building height (Fig. 1).

ICESat data were also collected for evaluating the estimated building height. The GLAS sensor on the ICESat receives waveform laser returns over ground footprints that are approximately 70 m circles in diameter (Cheng et al., 2011). We used the GLA14 product from ICESat because it contains parameters of waveform Gaussian fitting of laser returns (Simard et al., 2011). Thus, the relative height of detected target by ICESat can be measured as the distance between the first (e.g., building) and the last (e.g., ground) returns (Fig. S1) (Gong et al., 2011; Sun et al., 2008). All ICESat footprints in major US cities were used in our evaluation, with a temporal span from 2003 to 2009.

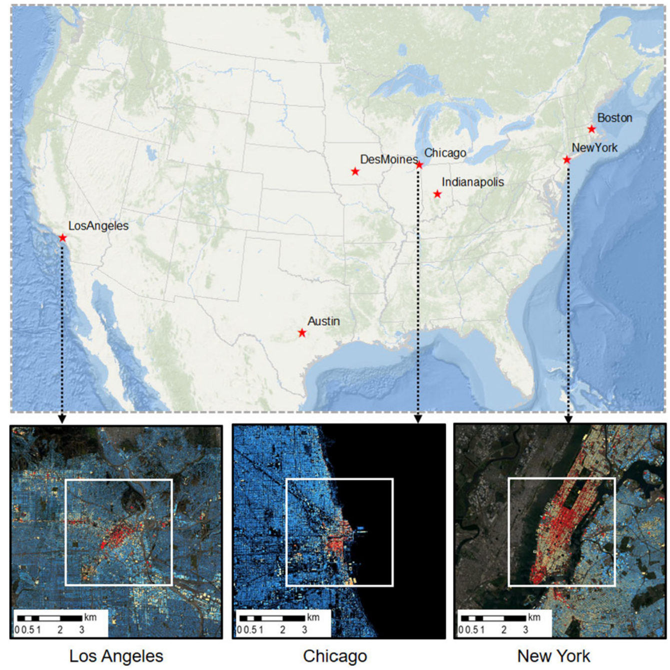

Fig. 1. Seven US cities with the reference building height data and three enlarged examples of LiDAR building height in Los Angeles, Chicago, and New York. The spatial extent of enlarged examples is 20 km × 20 km. The white frame presents the 10 km × 10 km extent in the model development.

The SRTM and ALOS data were used to estimate the height for comparison with our results in seven cities in the US (Fig. 1). The digital elevation model (DEM) from SRTM was measured by two SAR instruments on the space shuttle (Farr et al., 2007), and the digital surface model (DSM) from ALOS was calculated using triplet stereo images (Takaku et al., 2016). The difference between DSM and DEM in the urban area was regarded as the building height for comparison. It is worth to note that the SRTM was acquired in 2002 while the ALOS data were acquired from 2006 to 2011.

3. Method

In this study, we developed a method to estimate building height from the Sentinel-1 data and then calculated building height of major cities in the US (Fig. 2). We defined the resulting building height as the mean height estimated from Sentinel-1 data within the 500 m grid, including buildings and non-buildings such as streets and parking lots. First, we proposed a new indicator of VVH to better capture building height using backscatter coefficients at VV and VH polarizations from the Sentinel-1 GRD data (Fig. 2a). Second, we developed a building height model using the reference height from LiDAR and VVH. We first determined the parameter γ through a sensitivity analysis when calculating the indicator of VVH, and then calibrated the building height

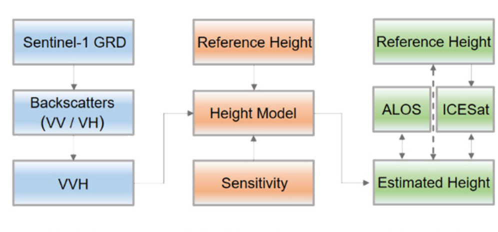

Fig. 2. The flowchart of building height estimation. The proposed indicator from Sentinel-1 GRD data (a). The development of building height model (b). The evaluation of generated building height (c).

model for the parameters of a, b, and c through a cross-validation (Fig. 2b). Third, we evaluated the estimated building height from Sentinel-1 data in seven US cities using the reference height and the ALOS derived height, and in US major cities using the ICESat data (Fig. 2c). Details of each step are presented in following sections.

3.1. Proposed indicator of VVH

After preprocessing Sentinel-1 GRD data, we proposed a new indicator for estimating building height. First, we aggregated backscatter coefficients of VV and VH from Sentinel-1 GRD data to a coarse resolution (e.g., 500 m). We retrieved mean backscatter coefficients at VV and VH polarizations in 2015 from Sentinel-1 GRD data, which have been preprocessed through noise removal, radiometric calibration, and terrain calibration (Ban et al., 2015). After that, we calculated the mean of high resolution backscatter coefficients (10 m) at a coarse resolution (e.g., 500 m) for the indicator of VVH. Similarly, we derived the mean height of the 500 m pixel with the consideration of high-resolution building height data from LiDAR. Thus, the aggregated mean backscatters and heights at a coarse resolution are linked for model development.

Second, we proposed an indicator from backscatter coefficients of VV and VH (Fig. 3). Backscatter coefficients from VV and VH show a good correspondence with the reference height data as for the case of the New York metropolitan area (Frolking et al., 2013) (Fig. 3a). That is, backscatter coefficients in high-rise building areas (e.g., the central business district (CBD)) are notably stronger than those in low-rise building areas (e.g., the residential district). However, the relationship between backscatter coefficient and building height is not consistent over different regions (Fig. 3b). For example, the building height in the region 1 is low, with high VV and low VH; whereas the building height in the region 3 is high, with high VV and VH. In region 2, the building height is medium, with medium VV and VH. In general, the building height is positively correlated with backscatter coefficients from Sentinel-1 data as reported in Koppel et al. (2017), whereas such a relationship varies with VV and VH. In previous studies, backscatter coefficients of VH were widely used for estimating forest biomass and vegetation height (Liao et al., 2018; Mermoz et al., 2015; Minh et al., 2016), suggesting that VH is a good proxy to characterize the vertical structure of vegetation. The capability of VH in reflecting the vertical information of land surfaces is also confirmed in our study as the result presented in Fig. S2. Nevertheless, in built environments with different layouts, structures, and materials of buildings, backscatter coefficients of VH and VV independently are inadequate to well capture heights in low- and high-rise buildings simultaneously (Fig. S2). This shortcoming can be mitigated by considering both VV and VH in the proposed new indicator of VVH as shown in Eq. (1). Comparing to other forms, the proposed VVH shows a better performance and a lower uncertainty range of estimated building heights (Fig. S2).

$$VVH = VV * \gamma^{VH} \tag{1}$$

where γ is a parameter to characterize the relative impact of VH to the derived VVH.

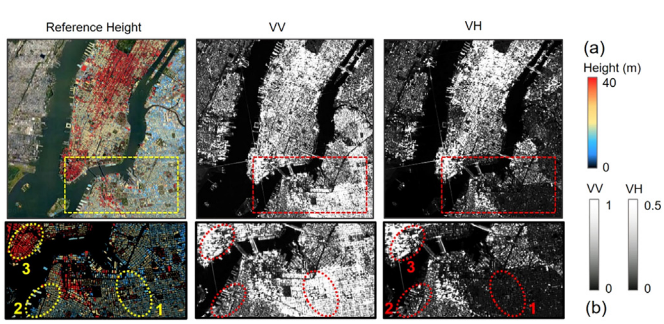

Fig. 3. Illustration of the relationship between building height and backscatter coefficients (VV and VH) from Sentinel-1 data in the New York metropolitan area (a). Enlarged views (b) of the rectangles in (a). The spatial extent of (a) is the same as the 10 km × 10 km extent in Fig. 1, with a spatial resolution of 10 m. Regions labeled as 1, 2, and 3 are representative cases with different heights and backscatter values (i.e., VV and VH).

3.2. Building height model

We developed the building height model using the reference height data and VVH in seven US cities, at an aggregated scale of 500 m (Fig. 4). It should be noted that the building height in this study specifically refers to the mean height within the 500 m grid, including buildings and non-buildings like streets and parking lots. In each city, we selected a 10 km × 10 km spatial extent centered on the CBD to ensure inclusion of high- and low-rise buildings with similar distributions among these seven cities. The model performance is not sensitive to spatial extents used, i.e., the result derived from 10 km × 10 km extent is similar as those from larger (e.g., 20 km × 20 km) and smaller (e.g., 5 km × 5 km) spatial extents (Fig. S3). Thereafter, we compared the derived relationship between building height and VVH at different resolutions from 100 m to 1000 m at a step of 100 m (Fig. 4a). Overall, there is a relatively robust relationship between building height and VVH across different resolutions. We characterized this relationship using the developed building height model as Eq. (2). The derived building height model shows better performance when the aggregation resolution is greater than 500 m, regarding the R^2 and the root mean square error (RMSE) (Fig. 4, b–c). To achieve a balance of model performance and spatial details of derived building heights, we chose 500 m as the aggregation resolution in this study.

$$\ln H = a * VVH^{b} + c \tag{2}$$

where H is the log-transformed building height; a, b, c are three parameters in the building height model (red line in Fig. 4). γ was initially set as e and was finalized after the sensitivity analysis.

We performed a sensitivity analysis of the derived building height model and estimated building height to the parameter γ in deriving VVH (Fig. 5). We selected New York City in this sensitivity analysis for its wide range of building height. γ is directly related to VVH, which further affects the derived relationship with the building height. Thus, we calculated building heights at different quantile levels (i.e., ranging from 5% to 100% at a step of 5%), and compared their relative gaps of the absolute height difference of estimated height to the reference using different γ. When γ is low, the contribution of VH is relatively weak, resulting in a notable underestimation of high-rise buildings (Fig. 5a). With the increase of γ, the model performance improves regarding R^2 and RMSE (i.e., R^2 increases from 0.73 to 0.90, and RMSE decreases from 0.84 to 0.51) (Fig. 5a). In addition, the derived building height at different quantile levels (Fig. 5b) becomes closer to that of LiDAR data when γ increases, especially for high-rise buildings. The mean relative gaps in different quantile levels reach the minimum when γ is about 5 (Fig. 5c). Therefore, γ was set as 5 for its robust performance (Fig. 5a) because the improvement of model performance is stabilized once the γ is larger than 5 and closer estimation of building height compared to the reference data in terms of relative gaps at different quantile levels (Fig. 5b and c).

Finally, we developed the proposed building height model through cross-validation, using reference height data (10 km × 10 km) in seven US cities. The parameter of γ was set as 5 and the aggregation scale is 500 m. The dataset of reference building height in seven US cities was divided into ten folds randomly, and the building height model was derived using nine folds for training and the remaining one for validation (Li et al., 2018a; Minh et al., 2016). Overall, the derived building height models perform well according to the cross-validation results (Fig. 6a). These ten models are close regarding the parameters of a, b, and c in Eq. (2). The estimated and reference building heights are

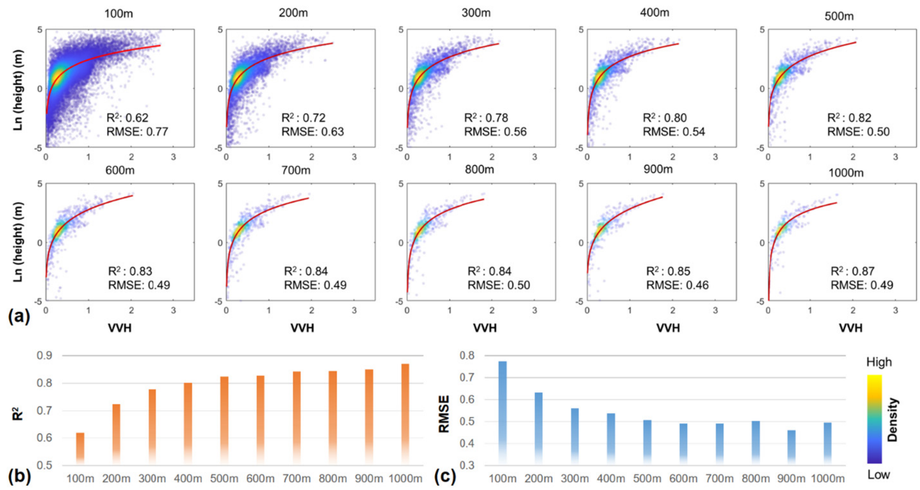

Fig. 4. Relationship between building height and VVH ($VV * e^{VH}$) in seven US cities at aggregated scales from 100 m to 1000 m at a step of 100 m (a) and their performances including R^2 (b) and RMSE (c). The R^2 and RMSE were calculated based on the log-transformed heights in Eq. (2).

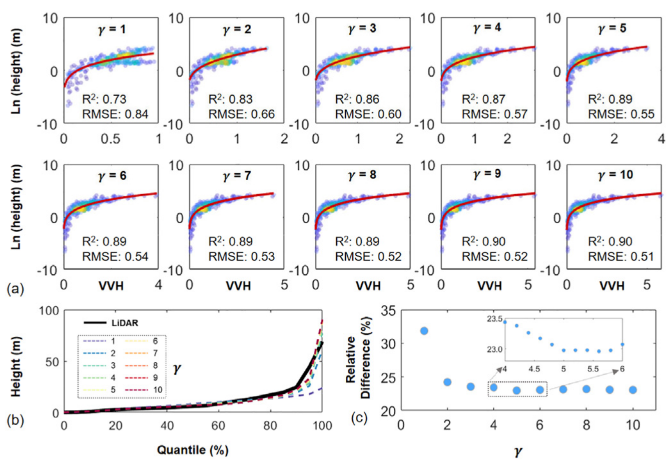

Fig. 5. Sensitivity of the derived building height model to γ at a 500 m resolution in New York City. The performance of building height model (red curve) under different γ (a). The estimated height at different quantile levels (dotted lines) under different γ (b). The relative difference of estimated to reference heights under different γ (c). The quantiles are values that partition the data of building heights into subsets (in total 100%) with an equal percentage at different levels (e.g., from 5% to 100% at a step of 5%). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

almost equally distributed around the 1:1 line (Fig. 6b), with a mean slope around 1.02. Their mean R^2 and RMSE are 0.78($\pm$ 0.04) and 0.49($\pm$ 0.03), respectively (Fig. 6b), suggesting that the building height models perform well. Although there are observations with larger biases in dotted ovals in Fig. 6, their proportion is less than 1%. Heights of these observations are low, and the biases are likely attributed to trees surrounding those low-rise buildings (Koppel et al., 2017). Given that the calibrated building height models in the cross-validation are close (Fig. 6a), we used their mean as the final model for estimating building height. The final parameters of *a*, *b*, and *c* are −23.61, −0.06, and 26.10, respectively.

3.3. Evaluation of estimated building height

We evaluated the estimated building height from Sentinel-1 data in two ways. First, we assessed our result using the ICESat data in all major cities in the US. Because the ICESat data measure the relative height within the footprint, the derive height is not directly comparable with the mean building height from the Sentinel-1 data. Therefore, we adjusted the ICESat height using the reference height from LiDAR images in seven cities in the US, and then evaluated the Sentinel-1 building height in major US cities using the adjusted ICESat height. Second, we evaluated the performance of building height models using all reference building height from LiDAR in seven US cities, and compared the estimated building height from Sentinel-1 data with that from the ALOS data.

4. Results and discussion

4.1. Comparison with ICESat height

The estimated building heights of major cities (in total 96) with area larger than 500 km^2 from Sentinel-1 data are comparable with that from ICESat data with a RMSE of 1.98 m (Fig. 7). The reference building height is 0.29 of the ICESat height in seven cities in the US (Fig. 7a). We adjusted the ICESat derived height (Fig. 7b) through multiplying all ICESat data with this factor of 0.29 (Fig. 7a). Then the estimated building heights from Sentinel-1 were compared with the adjusted

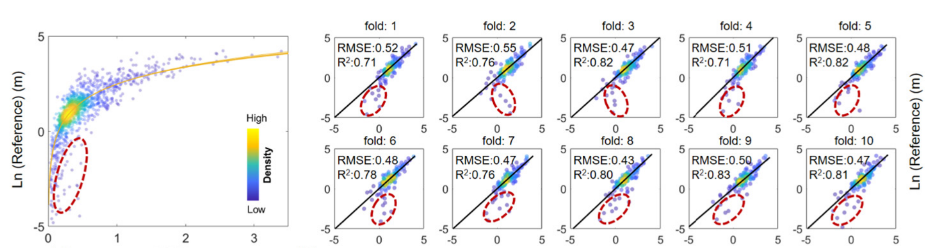

Fig. 6. Performance of the developed building height model at a 500 m resolution (a) in the cross-validation for each randomly divided fold (b). Observations in dotted ovals have relatively large biases.

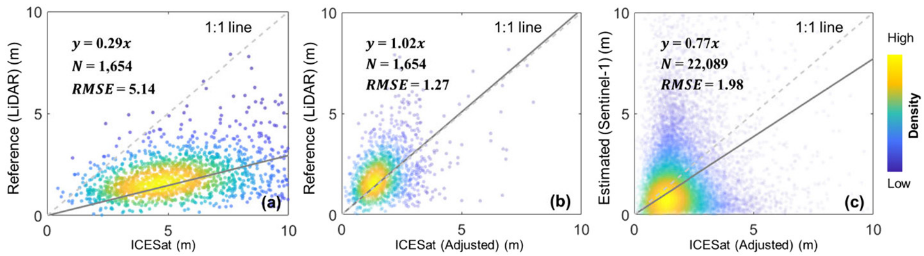

Fig. 7. Comparison between the reference LiDAR heights with ICESat heights (a) and the adjusted ICESat heights (b) in seven cities in the US. Comparison between the estimated height from Sentinel-1 data and the adjusted ICESat height (i.e., multiplying all ICESat data with the factor of 0.29) in major US cities (c). Dotted and solid lines are 1:1 line and the regressed line between two variables in x- and y-axes, respectively. N is the number of 500 m grids in urban areas.

ICESat height in major US cities, resulting in a RMSE of 1.98 m (Fig. 7c). There are larger discrepancies in higher buildings (black frame in Fig. S4a). These regions are mainly in highly developed areas (e.g., densely distributed buildings) (Fig. S4b), where the derived backscatter coefficients from Sentinel-1 data are likely amplified due to the double-bounce scattering effect (Li et al., 2016). Also, it is worth to note that ICESat derived heights are likely to be underestimated in these regions, because their heights could be calculated based on waveforms of high and low buildings (not the ground) in the footprint (~70 m circles in diameter) (Takaku et al., 2016).

4.2. Comparison with ALOS data

Overall, the estimated building heights from Sentinel-1 data show a good agreement with the reference data from LiDAR and outperform the ALOS in seven US cities. But it is worth to note that there are uncertainties in the SRTM DEM data in calculating height together with the ALOS DSM, and such uncertainties could reduce the performance of building heights from the ALOS data (Fig. 8). We removed those non-urban pixels using the impervious surface threshold of 20% as defined in the national land cover database (Homer et al., 2015). The RMSE between the estimated building heights from Sentinel-1 data and the reference heights is approximately 1.5 m, and the correlation coefficient of these two data sources is 0.77 (Fig. 8a), suggesting a good performance relative to that from ALOS DSM (Fig. 8b). Considering the uncertainty in the SRTM data in calculating the building height from the ALOS DSM, we measured the mean absolute percentage error (MAPE) of building heights to the reference. The MAPE in estimated height from Sentinel-1 is 44%, notably lower than that from ALOS of 116%. Regarding the spatial distribution of high and low buildings in seven cities, the estimated building heights from Sentinel-1 data also show a good agreement with the reference data (Fig. 9). The disagreement between building height from ALOS and the reference data is larger in some cities (e.g., Indianapolis and Los Angeles). Even in some areas, the ALOS building height is negative due to uncertainties in ALOS and SRTM. The whole city maps can be found in Fig. S5.

4.3. Uncertainty of derived building height data

There are associated uncertainties in the derived building height data from the definition of building height, backscatter coefficients from the Sentinel-1 data, and the proposed building height model. First, the building height defined in our result specifically refers to the mean height estimated from Sentinel-1 data within the 500 m grid, including buildings and non-buildings such as streets and parking lots. Thus, the derived building height is lower than that of a single building (Fig. 10). This definition can be applied to different resolutions according to specific applications (Fig. 4), although it is different from that of the building footprint.

Second, there are uncertainties in the reference height and the backscatter coefficients from Sentinel-1 data. Although the Sentinel-1 data were derived in the same year of 2015, the reference height data were produced in different years in seven cities, ranging from 2010 to 2014 (Table S1). Change of building heights (e.g., due to urban sprawl and reconstruction) during this period could be a source of uncertainties. Also, the derived backscatter coefficients from Sentinel-1 data contain uncertainties due to possible issues of double bouncing scattering caused by building structures and layout, as well as the high reflectivity from specific metal materials (e.g., containers in coastal areas) (Koppel et al., 2017; Li et al., 2016), although the total area of these regions is small relative to the total urban areas. In addition, the tree canopy coverage and the density of buildings can influence the estimated building height from the Sentinel-1 data. The higher canopy coverage and lower building density likely lead to a lower model performance (Table S2).

Third, although the proposed building height model can well capture variations in height under different VVHs, the uncertainties could vary from low-rise to high-rise buildings (Fig. S6). On the one hand, although heights of skyscrapers are notably higher than surrounding middle-rise buildings in CBD areas, the derived backscatter coefficients from Sentinel-1 data of these two building types could be similar. As a result, the estimated building height from Sentinel-1 data was underestimated for these skyscrapers. On the other hand, for some low-rise buildings with similar heights, their backscatter coefficients could be different due to ground layouts and materials of buildings, although our

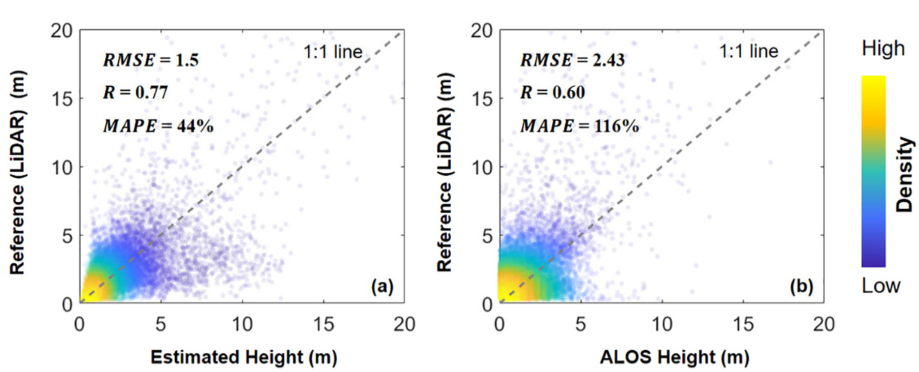

Fig. 8. Comparison of estimated building height at a 500 m resolution from Sentinel-1 (a) and ALOS (b) data with the reference heights from LiDAR data in seven US cities. Dotted line is the 1:1 line. MAPE: mean absolute percentage error.

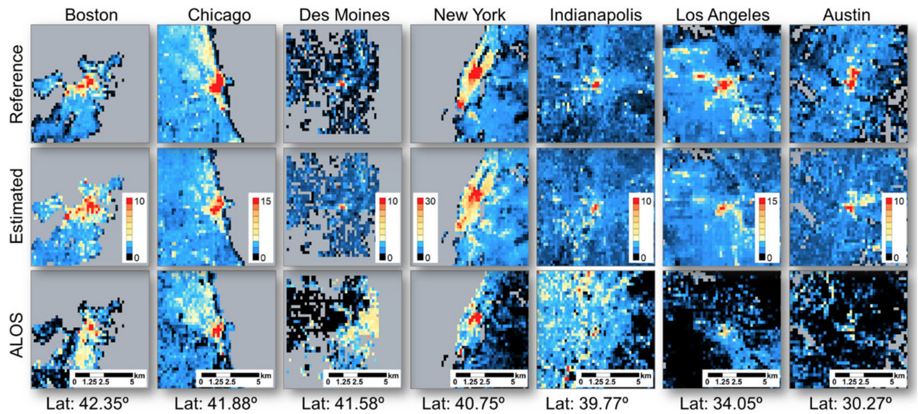

Fig. 9. Comparison of spatial patterns of building height (unit: m) at 500 m in seven US cities with a spatial extent of 20 km × 20 km. Areas outside the study domain are colored as grey.

building height model can capture heights of most buildings using the indicator of VVH.

4.4. Building height of major cities in the US

Overall, there are more cities with tall buildings in the Northeastern US compared to the Western and Southern US. These tall buildings commonly are in CBDs, and were used as a relative concept compared to those low-rise buildings in urban fringe areas (Fig. 10a). The mean height of each city (dots in Fig. 10a) was calculated within the urban boundary (Fig. S7), which was derived from nighttime light data (Zhou et al., 2018). Therefore, the mean height of each city is notably lower comparing to 500 m grid height. Large variations of building height among cities are related to various factors such as urban form, building structure, and population density (Koppel et al., 2017; Liao et al., 2018). For example, buildings in residential areas of selected cities in New York, Chicago, Atlanta, Los Angeles, Phoenix, and Dallas are notably different in terms of the mean height, lay out, and roof of buildings. Overall, building heights in New York and Chicago are higher compared to the other four cities, and New York has the largest building height variation in these six cities (Fig. 10b and Fig. S8). The vertical information of building heights can reflect the intensity of human activities and further help the identification of urban function types (e.g., residential and commercial areas) (Hu et al., 2016). The variation of building heights in cities also shows broad implications for studies such as urban climate, energy use, and greenhouse gas emissions. The 3-D structure of buildings can impact the urban climate (e.g., urban heat island and wind) in cities, and the information of urban height is helpful for improving urban climate modeling (Chen et al., 2011), particularly for cities in arid regions such as Los Angeles and Phoenix. Moreover, the new information of building heights can potentially improve the estimation and modeling of urban building energy use and greenhouse gas emissions (Güneralp et al., 2017; Xi et al., 2016). In addition, within each city, our Sentinel-1 based building heights capture a distinct attenuation of building heights from the CBD to surrounding residential areas as seen from the 3-D building maps (Fig. 10a). The derived building height data are of use for evaluating the change of population density across space and time because of their relationship and the nontrivial efforts in estimating gridded population density. Compared to other five cities among six example cities (Fig. 10), there are more tall buildings in the financial neighborhood of Manhattan in New York due to its commercial use with a high population density in daytime, and these tall buildings result in a high mean height.

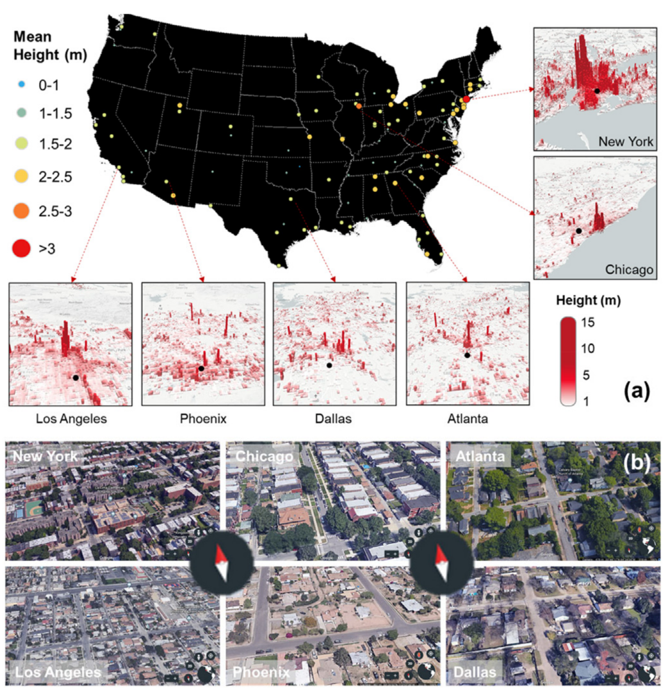

Fig. 10. Mean building heights of major US cities and 3-D building maps from Sentinel-1 data of six example cities (a) and enlarged views from Google Earth 3D map (b). The locations of the enlarged views are shown as black dots in (a). Snapshots in (b) were derived using Google Earth URL using the same parameters (e.g., compass and camera height). The spatial extent of each snapshot is about 160 m × 320 m.

5. Conclusions

In this study, we developed a method to estimate building height from the Sentinel-1 data and calculated building height of major cities in the US at a 500-m resolution. Overall, results from our model show a good agreement with the reference building height from LiDAR images based on the leave-one-out cross-validation in the CBD surrounded area of seven US cities. The RMSE between the estimated and reference height is 1.5 m for entire regions of seven cities in the US, which is better compared to building height from ALOS data. Also, our estimated heights are rational at the national scale, through comparing with ICESat results in major cities of the US. Our findings show large variations in building height among cities in the US. We find that building heights for cities in the Northeastern US are higher compared to the Western and Southern US. In addition, there are more high-rise buildings in the CBD of large cities (Fig. 10). Such spatially explicit patterns of building heights are related to a variety of factors such as population density and economic activity (Frolking et al., 2013; Koppel et al., 2017; Mahendra and Seto, 2019).

The estimated building height in US cities is valuable in urban studies such as urban surface process and building energy use modeling. The approach developed in this study can be used to build a global estimate of height, which is helpful in modeling urban surface process by improving urban canopy parameters for their spatial heterogeneity (Jackson et al., 2010) and building energy consumption by improving the estimation of floorspace for its spatial pattern (Mahendra and Seto, 2019). Also, the 3-D height information can be integrated with 2-D urban extents to predict more reliable population distribution and model the vertical growth of urban expansion (Gong et al., 2020; Li et al., 2017b; Li et al., 2018b; Xu and Coors, 2012). In addition, through integrating with ancillary datasets such as building footprints, the height information in our resulting products can be further improved. Such an improved building height dataset is helpful to evaluate the impact of the built environment on UHI and air pollution with the consideration of urban climate conditions (e.g., wind) that are affected by the distribution and height of buildings (Sobstyl et al., 2018).

Author contributions

Y.Z. designed the work; X.L. performed the analysis; X.L. and Y.Z. drafted the paper; and P.G., K.C.S., and N.C. contributed to the interpretation of the results and to the writing of the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2020.111705.

References

Ban, Y., Jacob, A., Gamba, P., 2015. Spaceborne SAR data for global urban mapping at 30 m resolution using a robust urban extractor. ISPRS J. Photogramm. Remote Sens. 103, 28–37.
Bonczak, B., Kontokosta, C.E., 2019. Large-scale parameterization of 3D building morphology in complex urban landscapes using aerial LiDAR and city administrative data. Comput. Environ. Urban. Syst. 73, 126–142.
Cárdenas Rodríguez, M., Dupont-Courtade, L., Oueslati, W., 2016. Air pollution and urban structure linkages: evidence from European cities. Renew. Sust. Energ. Rev. 53, 1–9.
Chen, F., Kusaka, H., Bornstein, R., Ching, J., Grimmond, C., Grossman-Clarke, S., Loridan, T., Manning, K.W., Martilli, A., Miao, S., 2011. The integrated WRF/urban modelling system: development, evaluation, and applications to urban environmental problems. Int. J. Climatol. 31, 273–288.
Cheng, F., Wang, C., Wang, J., Tang, F., Xi, X., 2011. Trend analysis of building height and total floor space in Beijing, China using ICESat/GLAS data. Int. J. Remote Sens. 32, 8823–8835.
Clinton, N., Gong, P., 2013. MODIS detected surface urban heat islands and sinks: global locations and controls. Remote Sens. Environ. 134, 294–304.
Farr, T.G., Rosen, P.A., Caro, E., Crippen, R., Duren, R., Hensley, S., Kobrick, M., Paller, M., Rodriguez, E., Roth, L., 2007. The shuttle radar topography mission. Rev. Geophys. 45.
Frolking, S., Milliman, T., Seto, K.C., Friedl, M.A., 2013. A global fingerprint of macro-scale changes in urban structure from 1999 to 2009. Environ. Res. Lett. 8, 024004.
Gong, P., Sheng, Y., Biging, G., 2002. 3D model-based tree measurement from high-resolution aerial imagery. Photogramm. Eng. Remote. Sens. 68, 1203–1212.
Gong, P., Li, Z., Huang, H., Sun, G., Wang, L., 2011. ICEsat GLAS data for urban environment monitoring. IEEE Trans. Geosci. Remote Sens. 49, 1158–1172.
Gong, P., Li, X., Wang, J., Bai, Y., Chen, B., Hu, T., Liu, X., Xu, B., Yang, J., Zhang, W., Zhou, Y., 2020. Annual maps of global artificial impervious areas (GAIA) between 1985 and 2018. Remote Sens. Environ. 236, 111510.
Güneralp, B., Zhou, Y., Ürge-Vorsatz, D., Gupta, M., Yu, S., Patel, P.L., Fragkias, M., Li, X., Seto, K.C., 2017. Global scenarios of urban density and its impacts on building energy use through 2050. Proc. Natl. Acad. Sci. 114, 8945–8950.
Gurney, K.R., Mendoza, D.L., Zhou, Y., Fischer, M.L., Miller, C.C., Geethakumar, S., de la Rue du Can, S., 2009. High resolution fossil fuel combustion CO2 emission fluxes for the United States. Environmental science & technology 43, 5535–5541.
Hao, L., Zhang, Y., Cao, Z., 2016. Building extraction from stereo aerial images based on multi-layer line grouping with height constraint. In: Geoscience and Remote Sensing Symposium (IGARSS), 2016 IEEE International. IEEE, pp. 445–448.
Homer, C.G., Dewitz, J.A., Yang, L., Jin, S., Danielson, P., Xian, G., Coulston, J., Herold, N.D., Wickham, J., Megown, K., 2015. Completion of the 2011 National Land Cover Database for the conterminous United States-representing a decade of land cover change information. Photogramm. Eng. Remote. Sens. 81, 345–354.
Hu, T., Yang, J., Li, X., Gong, P., 2016. Mapping urban land use by using Landsat images and open social data. Remote Sens. 8, 151.
Hutyra, L.R., Duren, R., Gurney, K.R., Grimm, N., Kort, E.A., Larson, E., Shrestha, G., 2014. Urbanization and the carbon cycle: current capabilities and research outlook from the natural sciences perspective. Earth's Future 2, 473–495.
Jackson, T.L., Feddema, J.J., Oleson, K.W., Bonan, G.B., Bauer, J.T., 2010. Parameterization of urban characteristics for global climate modeling. Ann. Assoc. Am. Geogr. 100, 848–865.
Koppel, K., Zalite, K., Voormansik, K., Jagdhuber, T., 2017. Sensitivity of Sentinel-1 backscatter to characteristics of buildings. Int. J. Remote Sens. 38, 6298–6318.
Li, H., Li, Q., Wu, G., Chen, J., Liang, S., 2016. The impacts of building orientation on polarimetric orientation angle estimation and model-based decomposition for multilook polarimetric SAR data in urban areas. IEEE Trans. Geoscience and Remote Sensing 54, 5520–5532.
Li, W., Zhou, Y., Cetin, K., Eom, J., Wang, Y., Chen, G., Zhang, X., 2017a. Modeling urban building energy use: a review of modeling approaches and procedures. Energy 141, 2445–2457.
Li, X., Gong, P., 2016. Urban growth models: progress and perspective. Science Bulletin 61, 1637–1650.
Li, X., Zhou, Y., 2017. Urban mapping using DMSP/OLS stable night-time light: a review. Int. J. Remote Sens. 38, 1–17.
Li, X., Gong, P., Liang, L., 2015. A 30-year (1984–2013) record of annual urban dynamics of Beijing City derived from Landsat data. Remote Sens. Environ. 166, 78–90.
Li, X., Lu, H., Zhou, Y., Hu, T., Liang, L., Liu, X., Hu, G., Yu, L., 2017b. Exploring the performance of spatio-temporal assimilation in an urban cellular automata model. Int. J. Geogr. Inf. Sci. 31, 2195–2215.
Li, X., Zhou, Y., Asrar, G.R., Zhu, Z., 2018a. Developing a 1 km resolution daily air temperature dataset for urban and surrounding areas in the conterminous United States. Remote Sens. Environ. 215, 74–84.
Li, X., Zhou, Y., Zhu, Z., Liang, L., Yu, B., Cao, w, 2018b. Mapping annual urban dynamics (1985–2015) using time series of Landsat data. Remote Sens. Environ. 216, 674–683.
Liao, C., Wang, J., Shang, J., Huang, X., Liu, J., Huffman, T., 2018. Sensitivity study of Radarsat-2 polarimetric SAR to crop height and fractional vegetation cover of corn and wheat. Int. J. Remote Sens. 39, 1475–1490.
Mahendra, A., Seto, K., 2019. Upward and Outward Growth: Managing Urban Expansion for More Equitable Cities in the Global South. World Resources Institute, Washington, DC.
Mermoz, S., Réjou-Méchain, M., Villard, L., Le Toan, T., Rossi, V., Gourlet-Fleury, S., 2015. Decrease of L-band SAR backscatter with biomass of dense forests. Remote Sens. Environ. 159, 307–317.
Minh, D.H.T., Le Toan, T., Rocca, F., Tebaldini, S., Villard, L., Réjou-Méchain, M., Phillips, O.L., Feldpausch, T.R., Dubois-Fernandez, P., Scipal, K., 2016. SAR tomography for the retrieval of forest biomass and height: cross-validation at two tropical forest sites in French Guiana. Remote Sens. Environ. 175, 138–147.
Pesaresi, M., Ehrlich, D., Ferri, S., Florczyk, A., Freire, S., Halkia, M., Julea, A., Kemper, T., Soille, P., Syrris, V., 2016. Operating Procedure for the Production of the Global Human Settlement Layer from Landsat Data of the Epochs 1975, 1990, 2000, and 2014. Publications Office of the European Union, pp. 1–62.
Seto, K.C., Dhakal, S., Bigio, A., Blanco, H., Delgado, G.C., Dewar, D., Huang, L., Inaba, A., Kansal, A., Lwasa, S., 2014. Human settlements, infrastructure and spatial planning. In: Edenhofer, O., Pichs-Madruga, R., Sokona, Y., Farahani, E., Kadner, K.S., Seyboth, A.A., Baum, S.B.I., Eickemeier, P., Kriemann, B., Savolainen, J., Schlömer, S., von Stechow, C., T.Z, Minx, J.C. (Eds.), Climate Change 2014: Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change, Cambridge, United Kingdom and New York, NY, USA.
Simard, M., Pinto, N., Fisher, J.B., Baccini, A., 2011. Mapping forest canopy height globally with spaceborne lidar. Journal of Geophysical Research: Biogeosciences 116.
Sobstyl, J., Emig, T., Qomi, M.A., Ulm, F.-J., Pellenq, R.-M., 2018. Role of city texture in urban heat islands at nighttime. Phys. Rev. Lett. 120, 108701.
Sun, G., Ranson, K., Kimes, D., Blair, J., Kovacs, K., 2008. Forest vertical structure from GLAS: an evaluation using LVIS and SRTM data. Remote Sens. Environ. 112, 107–117.
Sun, Y., Shahzad, M., Zhu, X.X., 2017. Building height estimation in single SAR image using OSM building footprints. In: Urban Remote Sensing Event (JURSE), 2017 Joint. IEEE, pp. 1–4.
Takaku, J., Tadono, T., Tsutsui, K., Ichikawa, M., 2016. Validation of "AW3D" global Dsm generated from Alos prism. ISPRS Annals of the Photogrammetry, Remote Sensing

and Spatial Information Sciences 3, 25.
Thiele, A., Cadario, E., Schulz, K., Thonnessen, U., Soergel, U., 2007. Building recognition from multi-aspect high-resolution InSAR data in urban areas. IEEE Trans. Geosci. Remote Sens. 45, 3583–3593.
Torres, R., Snoeij, P., Geudtner, D., Bibby, D., Davidson, M., Attema, E., Potin, P., Rommen, B., Floury, N., Brown, M., 2012. GMES Sentinel-1 mission. Remote Sens. Environ. 120, 9–24.
Wang, W., Xu, Y., Ng, E., Raasch, S., 2018. Evaluation of satellite-derived building height extraction by CFD simulations: a case study of neighborhood-scale ventilation in Hong Kong. Landsc. Urban Plan. 170, 90–102.
Wang, X., Yu, X., Ling, F., 2014. Building heights estimation using ZY3 data—a case study of Shanghai, China. In: Geoscience and Remote Sensing Symposium (IGARSS), 2014 IEEE International. IEEE, pp. 1749–1752.
Xi, F., Davis, S.J., Ciais, P., Crawford-Brown, D., Guan, D., Pade, C., Shi, T., Syddall, M., Lv, J., Ji, L., 2016. Substantial global carbon uptake by cement carbonation. Nat. Geosci. 9, 880.
Xu, Y., Ma, P., Ng, E., Lin, H., 2015. Fusion of WorldView-2 stereo and multitemporal TerraSAR-X images for building height extraction in urban areas. IEEE Geosci. Remote Sens. Lett. 12, 1795–1799.
Xu, Z., Coors, V., 2012. Combining system dynamics model, GIS and 3D visualization in sustainability assessment of urban residential development. Build. Environ. 47, 272–287.
Yu, B., Liu, H., Wu, J., Hu, Y., Zhang, L., 2010. Automated derivation of urban building density information using airborne LiDAR data and object-based method. Landsc. Urban Plan. 98, 210–219.
Zhou, Y., Li, X., Asrar, G.R., Smith, S.J., Imhoff, M., 2018. A global record of annual urban dynamics (1992–2013) from nighttime lights. Remote Sens. Environ. 219, 206–220.